

Term Report

Submission date: 2026/07/18

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| Surname / Given name<br>(the same on passport) | Yadav / Rikesh                                                                                      |
| Your Job title                                 | Research Intern                                                                                     |
| Inst. Name of Internship                       | Yuan Ze University, Taoyuan, Taiwan                                                                 |
| Name of PI / Department / School               | Asst. Prof. Dr. Ihsan Ullah / INEC Laboratory /<br>Computer Science Engineering, Yuan Ze University |
| Date of Internship                             | 2026/06/09 to 2026/08/29                                                                            |

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| <div><div>The specific results and performance of internship</div><div>(within 1000 words, but PI can require more words if necessary)</div></div> | <p>I am currently completing a research internship at the Intelligent Networks and Edge-Cloud Computing (INEC) Laboratory, Yuan Ze University, Taiwan, under the supervision of Assistant Professor Dr. Ihsan Ullah, as part of the International Internship Pilot Program (IIPP). The internship spans from June 9, 2026 to August 29, 2026, following a structured 12-week plan designed to build expertise in Deep Reinforcement Learning and Graph Neural Networks, with the goal of developing a research prototype and contributing toward a future IEEE conference publication.</p> <p><b>Weeks 1–2: Reinforcement Learning and Deep Q-Network Implementation</b></p> <p>The first two weeks were dedicated to building a strong foundation in Reinforcement Learning (RL). I started by studying the core concepts of the Markov Decision Process framework — how an agent observes a state, takes an action, receives a reward, and transitions to a new state. The interaction cycle below illustrates this loop, which is the fundamental basis for all RL algorithms studied during this internship.</p> <div><div><p>The agent observes the current state <math>s_t</math>, takes an action <math>a_t</math>, receives reward <math>r_t</math>, and the environment transitions to the next state <math>s_{t+1}</math>.</p></div><div><p>The agent learns the optimal path from Start (S) to Goal (G) by updating the Q-table using the Bellman equation.</p></div></div> <p><i>Figure 1: RL Interaction Cycle agent-environment loop</i></p> <p><i>Figure 2: GridWorld 5×5 environment implemented from scratch in Python.</i></p> <p>To solidify this understanding practically, I implemented a GridWorld environment entirely from scratch in Python without any external RL library. The</p> |
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agent learned an optimal navigation policy through iterative Q-table updates governed by the Bellman equation, shown below:

$$Q(s, a) = Q(s, a) + \alpha \left[ r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Where:

- $Q(s, a)$  : Current Q-value of state  $s$  and action  $a$
- $\alpha$  : Learning rate ( $0 < \alpha \leq 1$ )
- $r$  : Reward received after taking action  $a$
- $\gamma$  : Discount factor ( $0 \leq \gamma < 1$ )
- $s'$  : Next state after taking action  $a$
- $\max_{a'} Q(s', a')$  : Maximum Q-value for next state  $s'$

This equation updates the Q-value towards the target value  $r + \gamma \max_{a'} Q(s', a')$ , enabling the agent to learn optimal policy.

Figure 3: Bellman Equation — the mathematical foundation of Q-Learning used in the GridWorld implementation.

In Week 2, I transitioned to Deep Q-Networks (DQN). After getting comfortable with PyTorch basics — tensors, nn.Module, loss functions, and backpropagation — I implemented a complete DQN agent to solve the CartPole-v1 environment from OpenAI Gymnasium. The DQN architecture and training workflow I built are shown below:

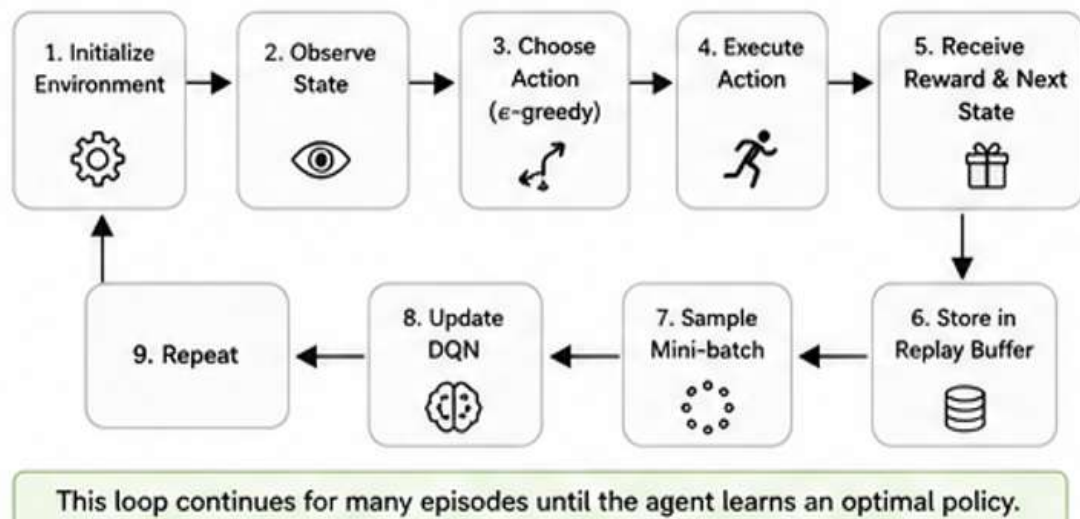


Figure 4: Complete DQN Training Workflow — 9-step loop from environment initialization to policy update.

A key innovation that stabilized my implementation was the Experience Replay Buffer, which stores past transitions and samples random mini-batches to break temporal correlations. The CartPole-v1 environment and the hyperparameters used are shown below:

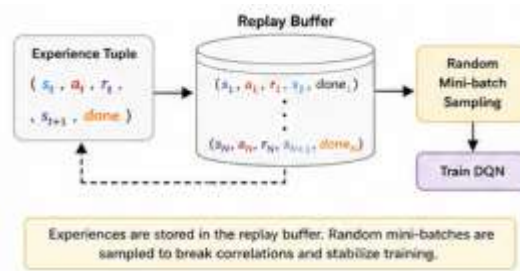


Figure 5: Experience Replay Buffer — stores  $(s, a, r, s', done)$  tuples for random mini-batch training

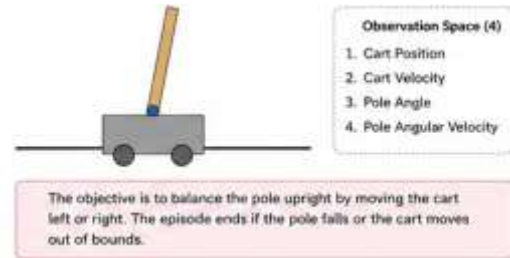


Figure 6: CartPole-v1 environment — 4-dimensional observation space, 2 discrete actions.

Through multiple systematic training runs with varying hyperparameters, the agent demonstrated a clear and consistent upward learning trend. The training performance and exact hyperparameters used are shown below:

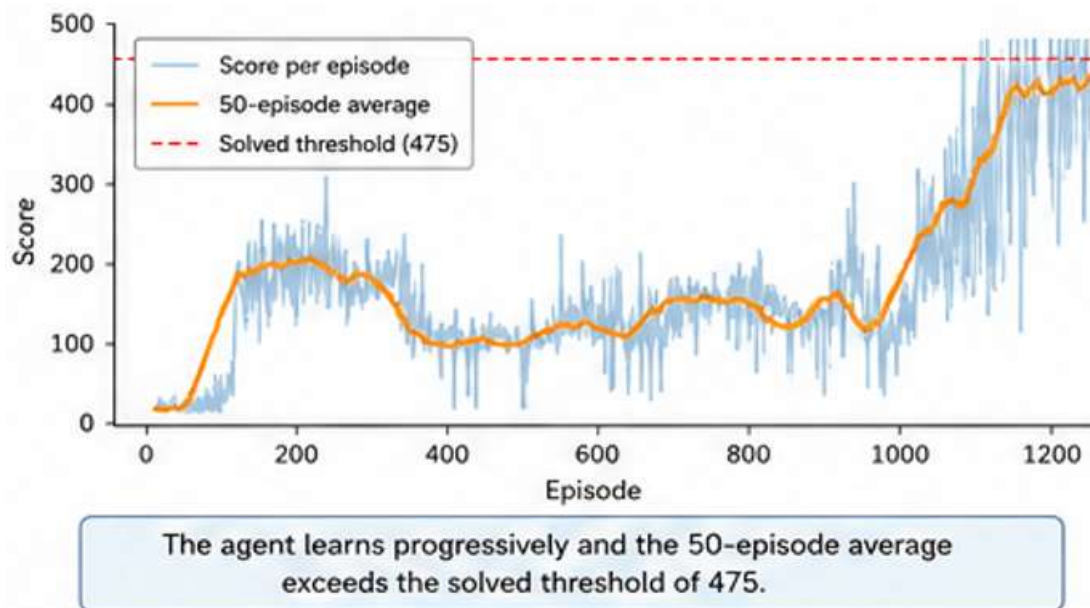


Figure 7: DQN training reward progression — 50-episode average approaching the solved threshold of 475.

All code was thoroughly documented with inline comments and module-level docstrings, then published to GitHub with a professional README covering concepts, architecture decisions, hyperparameter choices, and training results.

**GitHub - Q-Learning GridWorld:** <https://github.com/riekace/IIPP-RL-GCN-Research/tree/main/Week-1>

**GitHub DQN CartPole-v1:** <https://github.com/riekace/IIPP-RL-GCN-Research/tree/main/Week-2>

### Weeks 3–4: Advanced DQN Techniques and Literature Review

Weeks 3 and 4 deepened my understanding by studying three important DQN improvements from the research literature. Double DQN decouples action selection from value estimation to reduce overestimation bias. Dueling DQN separates Q-values into state value and action advantage streams. Prioritized Experience Replay samples more informative transitions more frequently. I extended experiments to the LunarLander-v2 environment, building my ability to design controlled experiments and compare results systematically.

In parallel, I conducted a focused literature review of 2024 and 2025 IEEE publications on GNN-based and DRL-based task offloading in edge computing.

Two key papers were analyzed: a 2024 study using GNNs and DQN as separate modules, and the 2025 GAPO paper using GNN-based Actor-Critic for vehicular edge computing. Both revealed consistent gaps pointing directly toward my research contribution.

### **Week 5: Graph Neural Networks and Research Problem Formulation**

Week 5 marked a meaningful shift. I began exploring Graph Neural Networks, specifically Graph Convolutional Networks (GCNs), studying how they represent graph-structured data using adjacency matrices and node feature matrices, and how message-passing enables nodes to aggregate neighborhood information across layers. I installed PyTorch Geometric (PyG) and implemented my first GCN for a node classification task, gaining hands-on graph learning experience before applying it to the research problem.

After in-depth discussions with Dr. Ihsan Ullah and evaluating multiple problem statements against existing literature, we finalized the research focus on developing an intelligent IoT sensor-based dynamic evacuation framework for smart environments — combining graph-based representations with reinforcement learning for adaptive decision-making using real-time sensor data.

*"Existing GNN-DRL approaches either treat GNN and DRL as separate modules or focus exclusively on vehicular environments. No prior work has proposed a unified GCN-DQN architecture for intelligent decision-making in stationary IoT environments with heterogeneous device profiles and real-time sensor integration."*

### **Results Expected and Plans for the Next Stage**

During the remaining weeks, I will build and evaluate the proposed GCN-DQN system: a simulation environment modeling buildings and evacuation routes as dynamic graphs, where the GCN encodes real-time sensor state and the DQN agent makes adaptive decisions under dynamic conditions such as blocked exits or changing crowd densities. I will validate through controlled experiments against clear baselines, build a functional research prototype, and prepare a research manuscript under Dr. Ihsan Ullah's guidance.

### **Other Interactions and Communications**

Throughout the internship, I have maintained regular communication with my PI, Dr. Ihsan Ullah, through weekly progress meetings presenting study outcomes, implementation results, and research planning updates. Progress reports have been prepared at the end of each major phase. I have also interacted with fellow IIPP interns at Yuan Ze University, creating a genuinely collaborative international research environment.

Outside the lab, living and studying in Taiwan has been a meaningful personal and professional experience — exposure to a new academic culture, independent research environment, and international collaboration that I look forward to carrying into my future graduate studies and beyond.